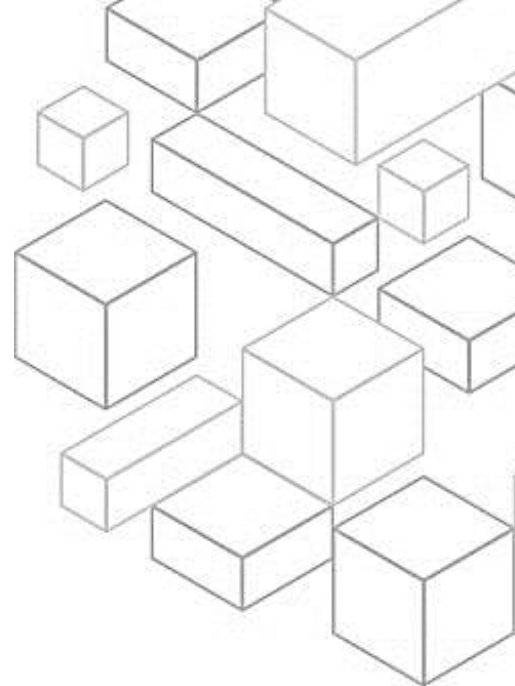




Module 9 AWS Well-Architected Sustainability

1.1 AWS Well-Architected

Welcome to module nine of AWS Well-Architected: Deep Dive on the Sustainability Pillar.

1.2 Learning objectives

In this module, you will learn about the sustainability pillar of the AWS Well-Architected Framework. You will also learn the design principles and best practices of the sustainability pillar.

1.3 Sustainability Pillar Overview

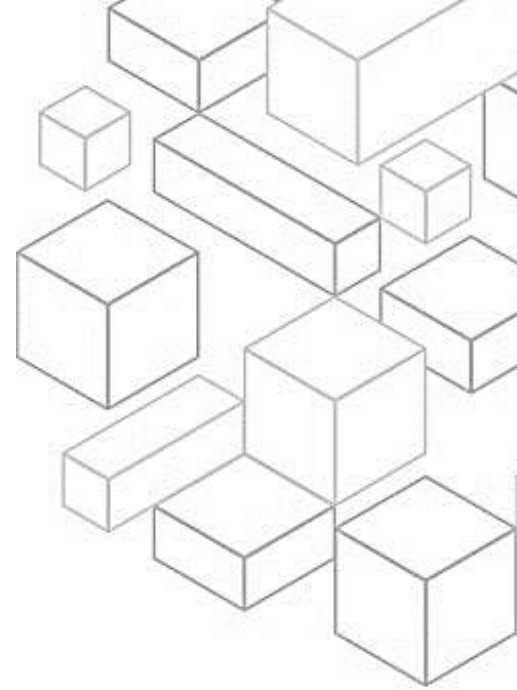
In this module, you will learn about the sustainability pillar of the Well-Architected Framework and get practical examples of the design principles using AWS architectural elements.

1.4 Pillars of AWS Well-Architected

There are six pillars in the Well-Architected Framework: operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability. These pillars are the foundations of the architecture for your technology solutions in the cloud.



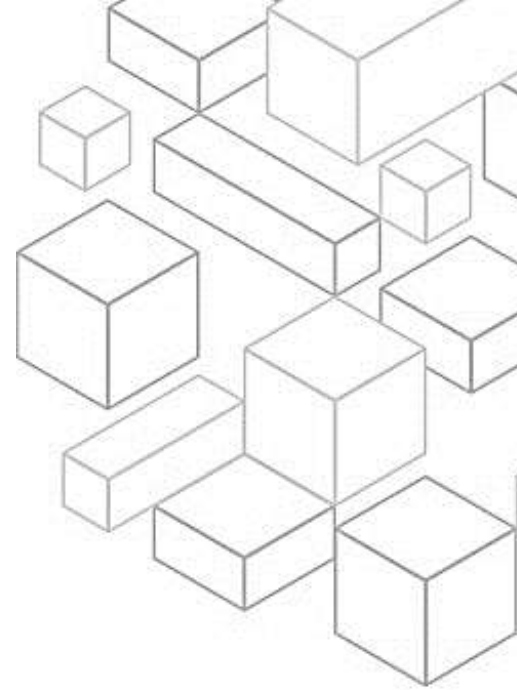
Sustainability is the most recent of the six pillars. It was introduced to the framework in 2021 to help customers make sustainability decisions about their workloads. This module will focus on the sustainability pillar.



1.5 What is the Sustainability pillar?

What is the sustainability pillar? The sustainability pillar enhances the framework to provide a way for you to consistently measure architectures against sustainability best practices and identify areas for improvement, with the focus being on reducing the energy consumption of your AWS workloads. It contains questions aimed at helping customers evaluate the design, architecture, and implementation of their workloads to reduce their energy consumption and improve their efficiency. Much more than a simple checklist, it's designed to be a tool that customers can use to track their progress toward policies and best practices that support a more sustainable future.

The pillar focuses on the sustainability best practices, which are understand, quantify, and apply. When building cloud workloads, the practice of sustainability is understanding the impacts of the services used, quantifying impacts through the entire workload lifecycle, and applying design principles and best practices to reduce these impacts. This module focuses on environmental impacts, especially energy consumption and efficiency, since they are important levers for architects to inform direct action to reduce resource usage.



When thinking about sustainability in your customers' architectures, it is important to remember that sustainability is a trade off just like many of the other AWS Well-Architected pillars. It is also important to understand that when it comes to improving architectures through sustainability, the shared responsibility model often used in discussing other pillars applies here as well. AWS is responsible for building a cloud infrastructure that is sustainable, and AWS customers are responsible for applying architectural best practices for sustainability in their cloud workloads.

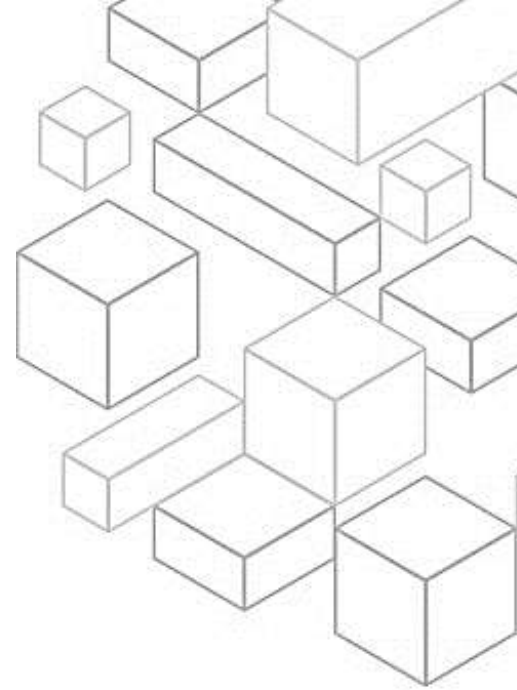
There are multiple reasons that sustainability can be an important consideration in improving architectures, including: customer demand, government regulations, employee demand, impact investing, and sustainability as competitive positioning

1.6 Sustainability

Now that you have a better understanding of the sustainability pillar, you can learn more about the components, starting with the sustainability design principles.

1.7 Sustainability design principles

There are six design principles for sustainability in the cloud.



The first design principle is to understand your impact. Measure the impact of your cloud workload and model future impact. Include all sources of impact, including those resulting from customer use of your products and those resulting from their eventual decommissioning and retirement. Compare the productive output with the total impact of your cloud workloads by reviewing the resources and emissions required per unit of work. Use this data to establish KPIs, evaluate ways to improve productivity while reducing impact, and estimate the impact of proposed changes over time.

Establish long-term sustainability goals, such as reducing the compute and storage resources required per transaction. Model the return on investment of sustainability improvements for existing workloads, and give owners the resources they need to invest in sustainability goals. You should also plan for growth and architect your workloads so that growth results in reduced impact intensity measured against an appropriate unit, such as per user or per transaction. Goals help you support the wider sustainability goals of your business or organization, identify regressions, and prioritize areas of potential improvement.

To maximize utilization, you can rightsize workloads and implement efficient design. This can help you ensure high utilization and maximize the energy efficiency of the underlying hardware. Two hosts running at 30 percent utilization are less efficient than one host running at 60 percent due to baseline power consumption per host. At the same time, eliminate or minimize

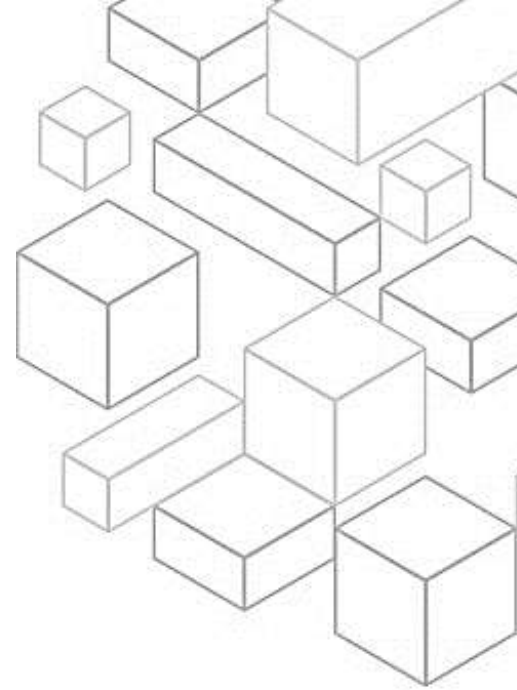


idle resources, processing, and storage to reduce the total energy required to power your workload.

You should also anticipate and adopt new, more efficient hardware and software offerings. Support the upstream improvements your partners and suppliers make to help you reduce the impact of your cloud workloads. You can continually monitor and evaluate your software offerings and design for flexibility to drive the rapid adoption of new efficient technologies.

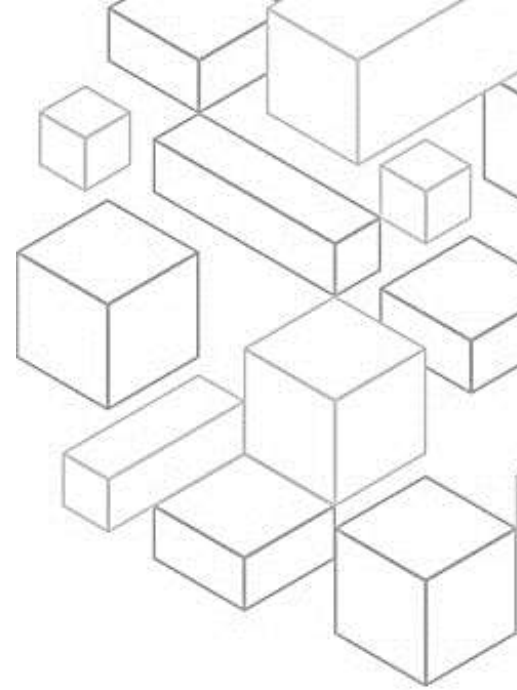
Another design principle is to use managed services. Sharing services across a broad customer base helps maximize resource utilization, which reduces the amount of infrastructure needed to support cloud workloads. For example, customers can share the impact of common data center components like power and networking by migrating workloads to the AWS Cloud and adopting managed services, such as AWS Fargate for serverless containers, where AWS operates at scale and is responsible for their efficient operation. Use managed services that can help minimize your impact, such as automatically moving infrequently accessed data to cold storage with Amazon S3 Lifecycle configurations or Amazon EC2 Auto Scaling to adjust capacity to meet demand.

Reduce the downstream impact of your cloud workloads by decreasing the amount of energy or resources required to use your services. Ease or eliminate the need for customers to upgrade their devices to use your services. You can





test using device farms to understand the expected impact and test with customers to understand the actual impact from using your services.



1.8 Sustainability

Now that you understand the sustainability design principles, you will dive even deeper into the sustainability best practices.

1.9 Sustainability best practice areas

In addition to design principles, there are also six best practice areas to focus on when working to implement sustainability in the cloud. These best practice areas are Region selection; alignment to demand; software and architecture patterns; data patterns; hardware and services; and process and culture.

In the remainder of this module, you will dive deeper into each of these best practice areas.

1.10 Region Selection

Region selection is the first sustainability best practice area you will explore.

1.11 Region selection

The choice of Region for your workload significantly affects its KPIs, including performance, cost, and carbon footprint. To effectively improve these KPIs, you



should choose Regions for your workloads based on business requirements and sustainability goals.

1.12 Alignment to Demand

The next best practice area in sustainability you will learn about is alignment to demand.

1.13 Alignment to demand

The way users consume your workloads and other resources can help you identify improvements to meet sustainability goals. One way to do this is to scale infrastructure to continually match user load and ensure that only the minimum resources required to support users are deployed. Using the elasticity of the cloud, you can scale your infrastructure dynamically to match the supply of cloud resources to demand and avoid overprovisioning capacity in your workload. You can also align SLAs with sustainability goals. To do this, review and optimize workload SLAs based on your sustainability goals to minimize the resources required to support your workload while continuing to meet business needs.

Another way to consider alignment to demand is to stop creating and maintaining unused assets. Decommission unused assets in your workload to reduce the number of cloud resources required to support your demand and minimize waste. Optimize geographic placement of workloads by reducing the distance network traffic must travel and decreasing the total network resources required to support your workload. Optimize team member resources to minimize the environmental sustainability impact while



supporting their needs. Implement buffering or throttling to flatten the demand curve and reduce the provisioned capacity required for your workload.

1.14 Software and Architecture Patterns

The next best practice area in sustainability is software and architecture patterns.

1.15 Software and architecture patterns

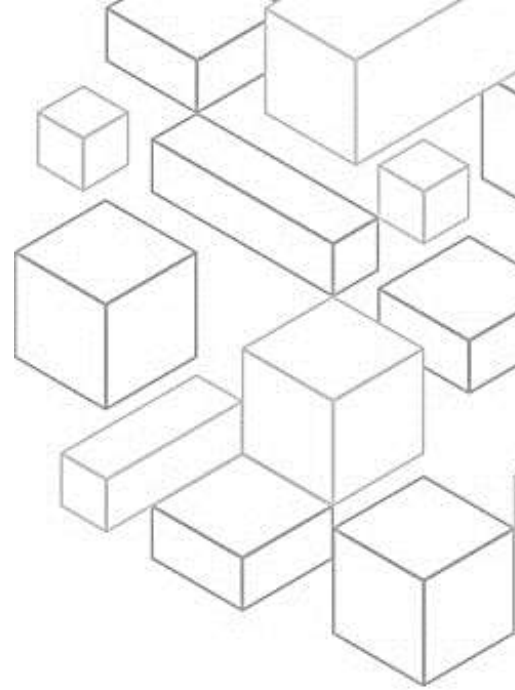
There are a few best practices for considering behavior patterns. First, optimize software and architecture for asynchronous and scheduled jobs. Use efficient software and architecture patterns such as queue-driven to maintain consistent, high utilization of deployed resources.

Another best practice is to remove or refactor workload components with low or no use. You can remove components that are unused and no longer required, and refactor components with little utilization to minimize waste in your workload. Also, optimize areas of code that consume the most time or resources. You can optimize code that runs within different components of your architecture to minimize resource usage while maximizing performance.

You can also optimize the impact on customer devices and equipment by understanding how they are used in your architecture and employing strategies to reduce their usage. This can minimize the overall environmental impact of your cloud workload. The last best practice is to use software patterns and



architectures that best support data access and storage patterns. Understand how data is used within your workload, consumed by your users, transferred, and stored. You can use software patterns and architectures that best support data access and storage to minimize the compute, networking, and storage resources required to support the workload.



1.16 Data Patterns

Now you will dive deeper into the data patterns best practice area in sustainability.

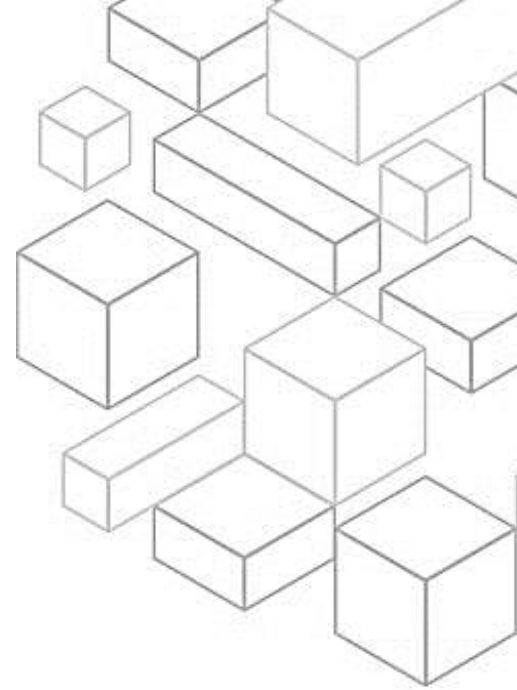
1.17 Data patterns

The first best practice for considering data patterns is to implement a data classification policy. You need to classify data to understand its criticality to business outcomes and choose the right energy-efficient storage tier to store the data. Also, use technologies that support data access and storage patterns. This can minimize the resources provisioned while supporting your workload.

Another best practice is to use policies to manage the lifecycle of your datasets and automatically enforce deletion timelines to minimize the total storage requirements of your workload. You can also use elasticity and automation to expand block storage or file system as data grows to minimize the total provisioned storage.



Another best practice is to remove unneeded or redundant data to minimize the storage resources required to store your datasets. Also, using shared file systems or storage can help you avoid data duplication and foster more efficient infrastructure for your workload.



You can minimize data movement across networks. Use shared file systems or object storage to access common data and minimize the total networking resources required to support data movement for your workload. Lastly, to minimize storage consumption, only back up data that has business value or is needed to satisfy compliance requirements. Examine backup policies and exclude ephemeral storage that doesn't provide value in a recovery scenario.

1.18 Hardware and Services

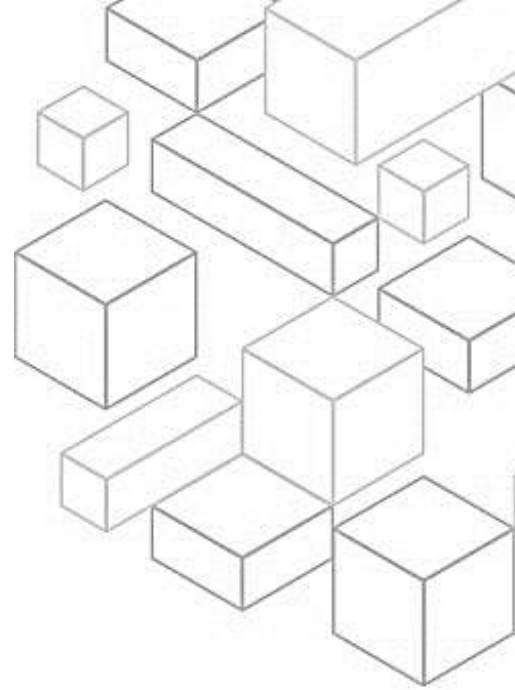
The next best practice area in sustainability is hardware and services.

1.19 Hardware and services

Look for opportunities to reduce workload sustainability impacts by making changes to your hardware management practices. Best practices for considering hardware patterns include using the minimum amount of hardware to efficiently meet your needs and using instance types with the least impact. Continually monitor and use new instance types to take advantage of energy efficiency improvements.



Another best practice is to use managed services to operate more efficiently in the cloud. Also, optimize your use of hardware-based compute accelerators to reduce the physical infrastructure demands of your workload.



1.20 Process and Culture

The last best practice area in sustainability you will explore is process and culture.

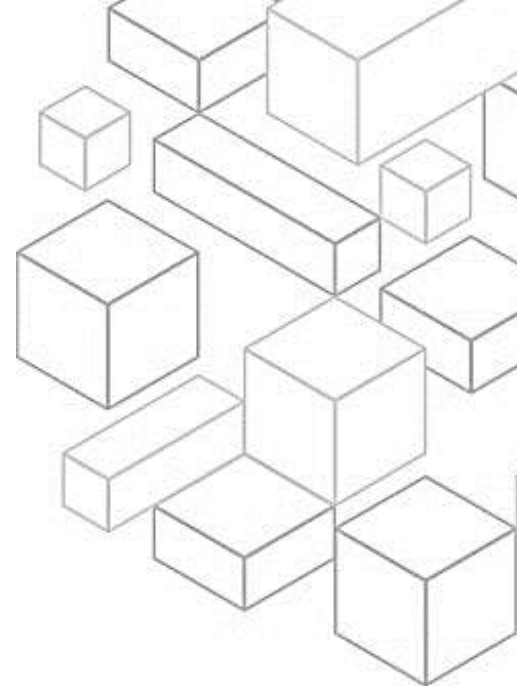
1.21 Process and culture

Look for opportunities to reduce your sustainability impact by making changes to your development, test, and deployment practices. Best practices include adopting methods and processes to validate potential improvements, minimize testing costs, and deliver small improvements.

You should also keep your workload up-to-date to adopt efficient features, remove issues, and improve the overall efficiency of your workload. Another best practice is to increase utilization of resources to develop, test, and build your workloads. And finally, use managed device farms to efficiently test a new feature on a representative set of hardware.

1.22 Question 1

1.23 Question 2



1.24 Question 3

1.25 Summary

In this module you learned about the sustainability pillar. This started with an overview and included in depth discussion on the value proposition, design principles, and best practices of the sustainability pillar.

1.26 Thank you

Thank you for participating in this deep dive on the sustainability pillar.

2. Keyboard Shortcuts

2.1 Keyboard Shortcuts